

Day 01

From ML apps to AI apps

What changed when we moved from training our own models to building products on top of large pretrained models.

Old vs new apps

AI vs ML vs DL

The ML workflow

How RAG works

Where AI is used

what changed?

Traditional ML app vs modern AI app

Traditional ML app

MODEL	You train your own, one per task
INPUT	Structured, tabular data (rows + columns)
OUTPUT	A number or a class: price, churn yes/no, fraud score
HARD PART	Collecting, cleaning and labelling data
EXAMPLES	Spam filter, credit scoring, demand forecast

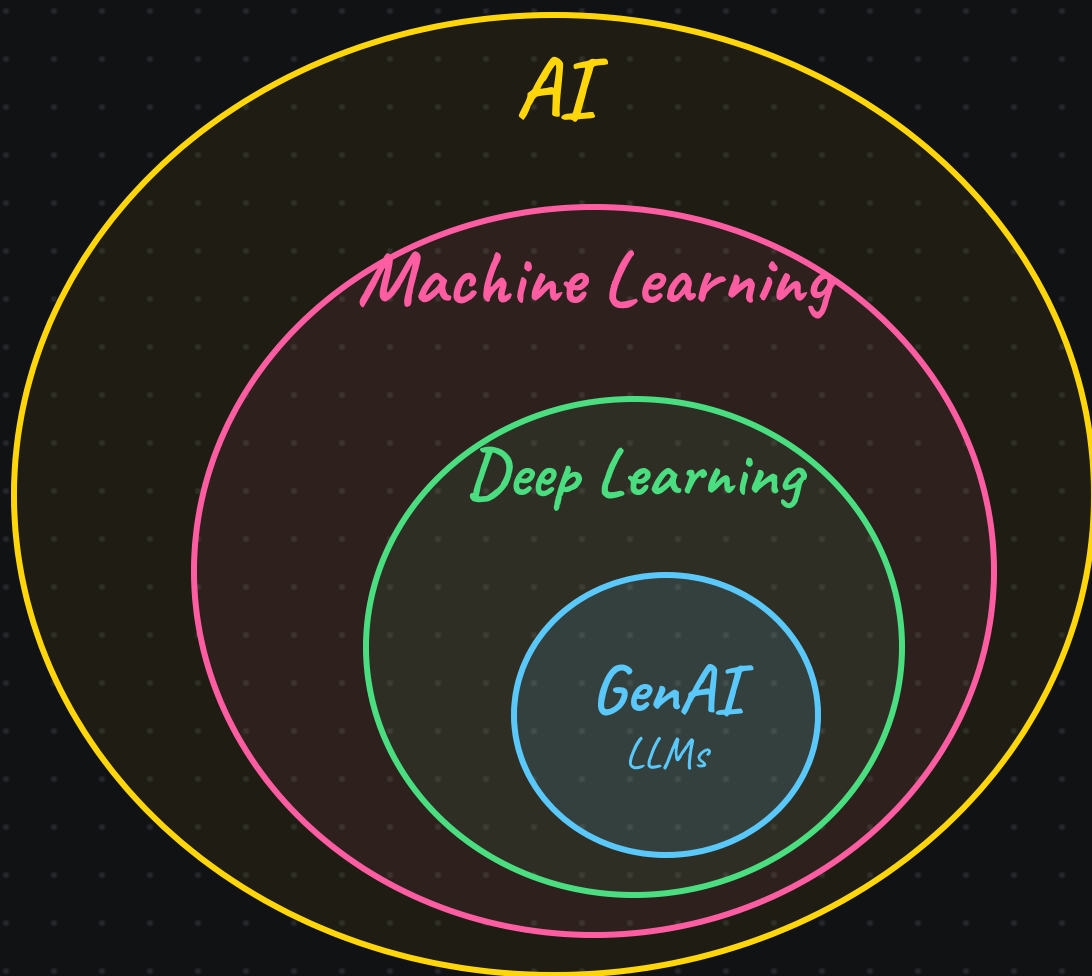
Modern AI app

MODEL	Call a pretrained LLM through an API; one model, many tasks
INPUT	Text, PDFs, images, audio
OUTPUT	Text, JSON, or actions (tool calls)
HARD PART	Giving it the right context, tools and tests
EXAMPLES	Chat with docs, AI agents, voice bots, extraction

The job moves from **training a model** → *building a system around one*

how they fit together

AI vs ML vs DL vs GenAI



AI

Any system that does a task we think needs intelligence. Can be plain hand-written rules.

Chess engine, rule-based chatbot

Machine Learning

Learns patterns from data instead of rules we write. Needs features picked by humans.

Linear regression, random forest, XGBoost

Deep Learning

ML with many-layered neural networks. Learns the features itself from raw images, audio, text.

CNNs, RNNs, Transformers

Generative AI

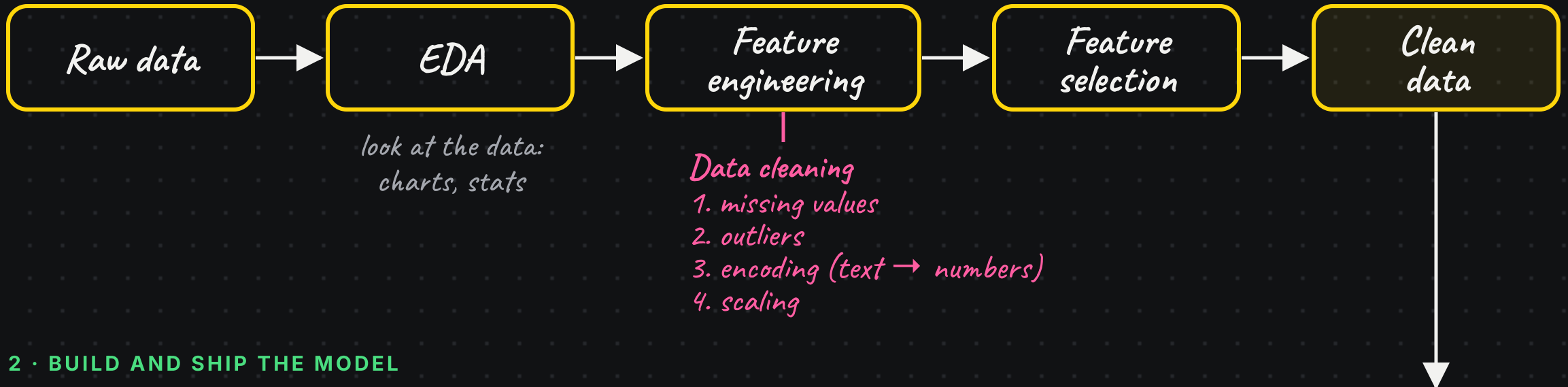
Deep learning models that create new content: text, code, images, speech.

GPT, Claude, Gemini, Llama

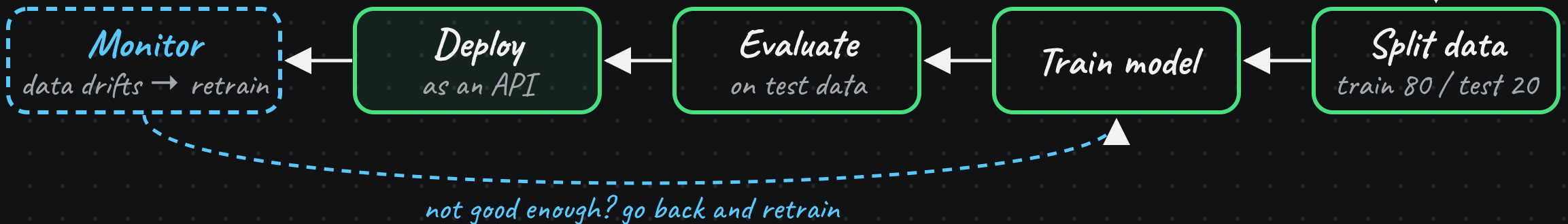
the old way

Traditional ML workflow

1 · PREPARE THE DATA



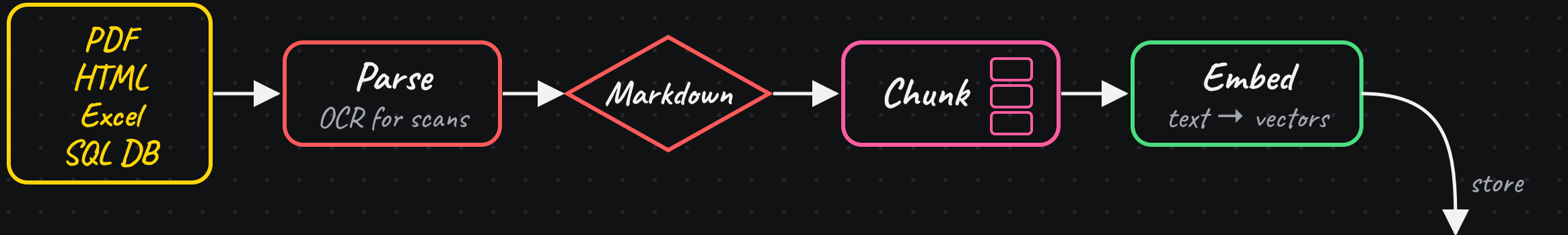
2 · BUILD AND SHIP THE MODEL



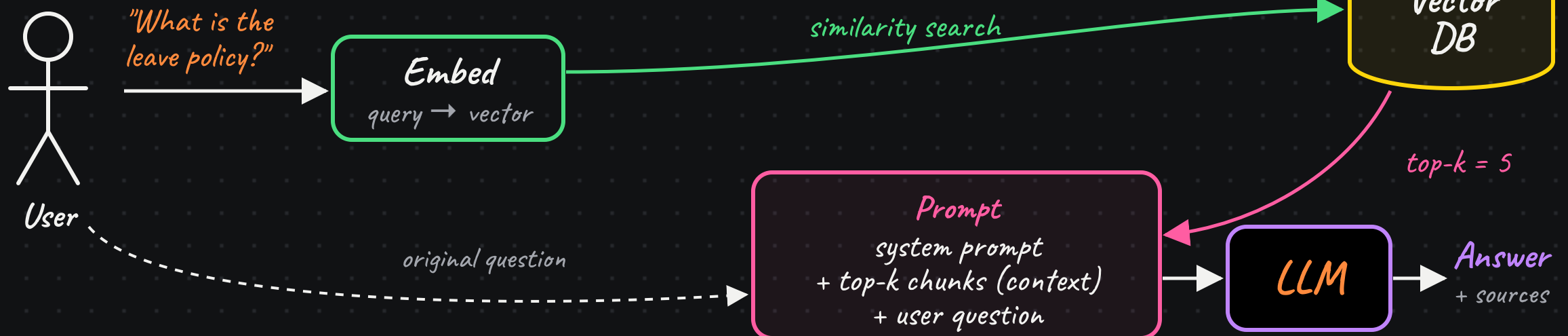
the new way, example #1

RAG: Retrieval-Augmented Generation

1 · INGESTION: RUNS ONCE PER DOCUMENT



2 · QUERY: RUNS ON EVERY QUESTION



what companies build today

Modern AI applications

RAG

The LLM answers from **your** data, not only what it learned in training.

e.g. chat with HR policies, product docs, a support knowledge base

AI agents

An LLM that plans, calls tools (APIs, DB, search) in a loop, and finishes a task.

e.g. research a topic, update a CRM, process a refund

Evals

Tests for AI output. A fixed set of questions, scored every time you change a prompt or model.

e.g. 50 questions checked for correct answer and correct source

Document extraction

Turn PDFs and scans into clean, structured JSON your app can save.

e.g. invoices, ID / KYC forms, contracts → fields in a database

Voice AI

Speech → text → LLM → speech, in real time, on a phone call or in an app.

e.g. support lines, booking appointments, lead qualification calls

remember this

Key takeaways

- 1 GenAI sits inside deep learning, inside ML, inside AI. LLMs are one kind of GenAI model.
- 2 Old apps: **train a model per task** on clean tabular data. Most of the work is data prep.
- 3 New apps: **use a pretrained LLM** and build the system around it. Context, tools, tests.
- 4 RAG = **find the right chunks, put them in the prompt**, let the LLM answer.

next class

Day 02 · How LLMs actually work